Sonning moduli. Modul qatnashgan tenglama va tengsizliklar

2-test (qiyinroq) · 20 ta savol

Ism-familiya: _____ Sinf: _____ Sana: _____

Har savolda bitta to'g'ri javobni belgilang.

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1. $|3x-4|=8$ **tenglamani yeching.**

A) -4 va $1\frac{1}{3}$

B) 4 va $-1\frac{1}{3}$

C) faqat 4

D) 4 va $1\frac{1}{3}$

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2. $|5-2x|=9$ **tenglamani yeching.**

A) faqat -2 **B)** 2 va -7 **C)** -2 va 7 **D)** -2 va -7

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3. $\left|\frac{1}{2}x-1\right|=3$ **tenglamani yeching.**

A) 4 va -2 **B)** 8 va 4 **C)** faqat 8 **D)** 8 va -4

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4. $|0{,}4x+1{,}2|=2$ **tenglamani yeching.**

A) 5 va -5 **B)** 2 va 8 **C)** -2 va 8 **D)** 2 va -8

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5. $|2x-3|<7$ **tengsizlikni yeching.**

A) $-5<x<2$

B) $-2<x<5$

C) $x<-2$ yoki $x>5$

D) $x<5$

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6. $|6-3x|\le 7$ **tengsizlikni yeching.**

A) $-4\frac{1}{3}\le x\le\frac{1}{3}$

B) $-\frac{1}{3}<x<4\frac{1}{3}$

C) $x\le-\frac{1}{3}$ yoki $x\ge4\frac{1}{3}$

D) $-\frac{1}{3}\le x\le4\frac{1}{3}$

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7. $|x+1|\ge 3$ **tengsizlikni yeching.**

A) $x\le-4$ yoki $x\ge2$

B) $x\le-2$ yoki $x\ge4$

C) $-4\le x\le2$

D) faqat $x\ge2$

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8. $|2x+5|>1$ **tengsizlikni yeching.**

A) $-3<x<-2$

B) $x<2$ yoki $x>3$

C) $x\le-3$ yoki $x\ge-2$

D) $x<-3$ yoki $x>-2$

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9. **Rasmdagi yechimlar to'plami qaysi tengsizlikka mos?**



A) $|x+4|\le3$

B) $|x-4|<3$

C) $|x-3|\le4$

D) $|x-4|\le3$

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10. **Rasmdagi yechimlar to'plami qaysi tengsizlikka mos?**



A) $|x+2| \le 3$ **B)** $|x+2| > 3$

C) $|x-2| \ge 3$ **D)** $|x+2| \ge 3$

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11. $|2x-1| < 5$ **tengsizlikning nechta butun yechimi bor?**

A) 3 **B)** 6 **C)** 5 **D)** 4

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12. $|3-x| \le 2$ **tengsizlikning eng katta butun yechimini toping.**

A) -1 **B)** 2 **C)** 5 **D)** 4

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13. $|2x-7|+3 \le 3$ **tengsizlikni yeching.**

A) yechim yo'q **B)** $x \le 3{,}5$

C) $x = 3{,}5$ **D)** istalgan son

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14. $|x+4|+6 \ge 2$ **tengsizlikni yeching.**

A) $x \le -8$ yoki $x \ge 0$ **B)** istalgan son

C) $x \ge -4$ **D)** yechim yo'q

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15. $a < 0$ **bo'lsa,** $|a| - a$ **ifoda nimaga teng?**

A) $2a$ **B)** $-2a$ **C)** $-a$ **D)** 0

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16. **Detalning uzunligi** l **(mm) uchun talab:** $|l-50| \le 0{,}2$**. Qaysi detal yaroqsiz?**

A) 50,25 mm **B)** 49,8 mm **C)** 49,95 mm **D)** 50,2 mm

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17. **Sut tashiydigan mashinada harorat** t **(°C) uchun talab:** $|t-4| \le 2$**. Harorat qaysi oraliqda bo'lishi kerak?**

A) 0 °C dan 8 °C gacha **B)** −2 °C dan 6 °C gacha

C) 2 °C dan 6 °C gacha **D)** 4 °C dan 6 °C gacha

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18. $|x-1| < 4$ **tengsizlikning nechta natural yechimi bor?**

A) 4 **B)** 7 **C)** 3 **D)** 5

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19. **Qaysi son** $|3x+2| \ge 4$ **tengsizlikning yechimi emas?**

A) -2 **B)** 1 **C)** -3 **D)** 0

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20. $|6-4x| = 10$ **tenglama ildizlarining yig'indisini toping.**

A) 3 **B)** -3 **C)** -1 **D)** 4

Javoblar

Sonning moduli. Modul qatnashgan tenglama va tengsizliklar · 2-test (qiyinroq)

№	Javob	Qiymati
1	**B**	4 va $-1\frac{1}{3}$
2	**C**	-2 va 7
3	**D**	8 va -4
4	**D**	2 va -8
5	**B**	$-2 < x < 5$
6	**D**	$-\frac{1}{3} \le x \le 4\frac{1}{3}$
7	**A**	$x \le -4$ yoki $x \ge 2$
8	**D**	$x < -3$ yoki $x > -2$
9	**D**	$\lvert x-4\rvert \le 3$
10	**D**	$\lvert x+2\rvert \ge 3$

№	Javob	Qiymati
11	**D**	4
12	**C**	5
13	**C**	$x = 3{,}5$
14	**B**	istalgan son
15	**B**	$-2a$
16	**A**	50,25 mm
17	**C**	2 °C dan 6 °C gacha
18	**A**	4
19	**D**	0
20	**A**	3

Qisqa yechimlar

1. $3x - 4 = 8$: $3x = 12$, $x = 4$.

$3x - 4 = -8$: $3x = -4$, $x = -1\frac{1}{3}$.

Javob: **4 va** $-1\frac{1}{3}$.

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2. $5 - 2x = 9$: $-2x = 4$, $x = -2$.

$5 - 2x = -9$: $-2x = -14$, $x = 7$.

Javob: -2 **va** 7.

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3. $\frac{1}{2}x - 1 = 3$: $\frac{1}{2}x = 4$, $x = 8$.

$\frac{1}{2}x - 1 = -3$: $\frac{1}{2}x = -2$, $x = -4$.

Javob: **8 va** -4.

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4. $0{,}4x + 1{,}2 = 2$: $0{,}4x = 0{,}8$, $x = 2$.

$0{,}4x + 1{,}2 = -2$: $0{,}4x = -3{,}2$, $x = -8$.

Javob: **2 va** -8.

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5. $-7 < 2x - 3 < 7$.

3 ni qo'shamiz: $-4 < 2x < 10$.

2 ga bo'lamiz: $-2 < x < 5$.

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6. $-7 \le 6 - 3x \le 7$.

6 ni ayiramiz: $-13 \le -3x \le 1$.

-3 ga bo'lamiz, belgilar teskari: $4\frac{1}{3} \ge x \ge -\frac{1}{3}$, ya'ni $-\frac{1}{3} \le x \le 4\frac{1}{3}$.

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7. $x + 1 \ge 3$: $x \ge 2$.

$x + 1 \le -3$: $x \le -4$.

Javob: $x \le -4$ **yoki** $x \ge 2$. (Ikkinchi holatni unutmang!)

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8. $2x + 5 > 1$: $2x > -4$, $x > -2$.
$2x + 5 < -1$: $2x < -6$, $x < -3$.
Belgi qat'iy: $x < -3$ **yoki** $x > -2$.

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9. Kesmaning o'rtasi: $(1 + 7) : 2 = 4$, o'rtadan chetlarigacha 3 birlik.
4 dan 3 birlikdan uzoq bo'lmagan nuqtalar, uchlari bo'yalgan: $|x - 4| \le 3$.

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10. -5 va 1 ning o'rtasi: $(-5 + 1) : 2 = -2$, undan chetlargacha 3 birlik.
-2 dan kamida 3 birlik uzoq nuqtalar: $|x - (-2)| \ge 3$, ya'ni $|x + 2| \ge 3$.

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11. $-5 < 2x - 1 < 5$, $-4 < 2x < 6$, $-2 < x < 3$.
Butun sonlar: $-1;\ 0;\ 1;\ 2$ **— 4 ta** (-2 va 3 kirmaydi).

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12. $-2 \le 3 - x \le 2$, $-5 \le -x \le -1$.
-1 ga bo'lamiz, belgilar teskari: $1 \le x \le 5$.
Eng katta butun yechim: **5**.

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13. 3 ni o'ng tomonga o'tkazamiz: $|2x - 7| \le 0$.
Modul manfiy emas, demak $2x - 7 = 0$: $x = 3{,}5$ **—** yagona yechim.

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14. 6 ni o'ng tomonga o'tkazamiz: $|x + 4| \ge -4$.
Modul doim ≥ 0, demak -4 dan katta: **istalgan son**.

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15. $a < 0$ da $|a| = -a$.
$|a| - a = -a - a = -2a$. Masalan, $a = -3$: $3 + 3 = 6 = -2 \cdot (-3)$.

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16. $-0{,}2 \le l - 50 \le 0{,}2$, ya'ni $49{,}8 \le l \le 50{,}2$.
49,8 va 50,2 **—** chegara, ular kiradi. Oraliqdan tashqarida: **50,25 mm**.

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17. $-2 \le t - 4 \le 2$.
4 ni qo'shamiz: $2 \le t \le 6$, ya'ni **2 °C dan 6 °C gacha**.

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18. $-4 < x - 1 < 4$, $-3 < x < 5$.
Natural sonlar: 1; 2; 3; 4 **— 4 ta**. (7 **—** barcha butun sonlar soni, $-2;\ -1;\ 0$ natural emas.)

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19. $x = 0$: $|2| = 2$, $2 < 4$ **—** bajarilmaydi.
$x = 1$: $|5| = 5$; $x = -2$: $|-4| = 4$; $x = -3$: $|-7| = 7$ **—** bajariladi. Javob: **0**.

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20. $6 - 4x = 10$: $x = -1$.
$6 - 4x = -10$: $x = 4$.
Yig'indi: $-1 + 4 = \mathbf{3}$.
